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Untitled2.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

C:\Users\user\OneDrive\Docu x + v
<global>
3. Issue Medicine
4. Search Medicine
5. Add New Medicine
6. Exit
Management
Projects
Workspace

Enter your choice: 1
MEDICINE STOCK MANAGEMENT REPORT

Medicine      Stock  DailyReq  MinStock  Expiry  Essential  Coverage  Condition  Priority
-----
Insulin        60    12        50        30      3          5.0      Expiry      55
Amoxicillin    200   25        80        20      3          8.0      Expiry      52
Oral Saline    120   35        80        45      3          3.4      Reorder     47
Paracetamol    300   40       100       120     2          7.5      Reorder     33
Antacid        250   18        60        15      1         13.9     Expiry      27
Cetirizine     180   15        50        90      1         12.0     Sufficient   8

Medicines are sorted according to management priority.

Report saved successfully in output.txt

1. Show Medicine Report
2. Add New Stock
3. Issue Medicine
4. Search Medicine
5. Add New Medicine
6. Exit

Enter your choice: |
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=====
EMERGENCY POTABLE WATER ALLOCATION
=====

1. Case 1 - Original Dataset
2. Case 2 - Enough Water
3. Case 3 - Below Minimum Requirement
4. Case 4 - Equal Priority
5. Case 5 - Increased Distribution Loss
6. Case 6 - Increased Waiting Period
7. Case 7 - Independent Test
8. Search a Zone
9. Find Highest Unresolved Requirement
10. Exit

Enter your choice: 2

===== CASE 2 =====
Enough Water for Complete Requested Quantity
Available Water: 22194.00 L

=====
ALLOCATION RESULT
=====
ID   Zone           Request  Minimum  Priority  Allocated  Delivered  Shortage  Status
-----
Z04  Ward 5          3800     1800     13.80    4318       3800       0         MINIMUM
Z03  Ward 3          4500     2000     13.20    4891       4500       0         FULL
Z01  Municipal Hospital 4000    3000     11.80    4082       4000       0         FULL
Z02  Central Flood Shelter 3500    2500     11.60    3646       3500       0         FULL
Z05  School Emergency Shelter 2600    1600     11.50    2737       2600       0         FULL
Z06  Fire and Emergency Service 2000    1500     10.90    2020       2000       0         FULL

=====

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water      : 20400.00 L
Total Allocated Water      : 21694.00 L
Total Effective Water Delivered : 20400.00 L
Total Unresolved Requirement : 0.00 L
Remaining Water            : 500.00 L

Service Conditions
-----
Full Requirement Satisfied : 5 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum              : 0 zone(s)
Completely Unserved        : 0 zone(s)
```

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Enter your choice: 3

===== CASE 3 =====
Enter available water in litres: 100

=====
ALLOCATION RESULT
=====
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
Z04 Ward 5 3800 1800 13.80 100 88 3712 BELOW MIN
Z03 Ward 3 4500 2000 13.20 0 0 4500 UNSERVED
Z01 Municipal Hospital 4000 3000 11.80 0 0 4000 UNSERVED
Z02 Central Flood Shelter 3500 2500 11.60 0 0 3500 UNSERVED
Z05 School Emergency Shelter 2600 1600 11.50 0 0 2600 UNSERVED
Z06 Fire and Emergency Service 2000 1500 10.90 0 0 2000 UNSERVED
=====

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water : 20400.00 L
Total Allocated Water : 100.00 L
Total Effective Water Delivered : 88.00 L
Total Unresolved Requirement : 20312.00 L
Remaining Water : 0.00 L

Service Conditions
-----
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum : 1 zone(s)
Completely Unservd : 5 zone(s)
```

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Z04 Ward 5 3800 1800 13.80 4318 3800 0 MINIMUM
Z03 Ward 3 4500 2000 13.20 4891 4500 0 FULL
Z01 Municipal Hospital 4000 3000 11.80 4082 4000 0 FULL
Z02 Central Flood Shelter 3500 2500 11.60 209 201 3299 BELOW MIN
Z05 School Emergency Shelter 2600 1600 11.50 0 0 2600 UNSERVED
Z06 Fire and Emergency Service 2000 1500 10.90 0 0 2000 UNSERVED
=====

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water : 20400.00 L
Total Allocated Water : 13500.00 L
Total Effective Water Delivered : 12500.53 L
Total Unresolved Requirement : 7899.47 L
Remaining Water : -0.00 L

Service Conditions
-----
Full Requirement Satisfied : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum : 1 zone(s)
Completely Unservd : 2 zone(s)

=====
EMERGENCY POTABLE WATER ALLOCATION
=====
1. Case 1 - Original Dataset
2. Case 2 - Enough Water
3. Case 3 - Below Minimum Requirement
4. Case 4 - Equal Priority
5. Case 5 - Increased Distribution Loss
6. Case 6 - Increased Waiting Period
7. Case 7 - Independent Test
8. Search a Zone
9. Find Highest Unresolved Requirement
10. Exit

Enter your choice: |
```

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Enter your choice: 5

===== CASE 5 =====
Residential zones are: Z03 and Z04
Enter residential Zone ID: Z03
Current distribution loss: 8.00%
Enter new distribution loss (%): 10

New loss of Z03 = 10.00%

=====
ALLOCATION RESULT
=====
ID   Zone           Request  Minimum  Priority  Allocated  Delivered  Shortage  Status
-----
Z04  Ward 5           3800     1800     13.80    4318      3800      0         MINIMUM
Z03  Ward 3           4500     2000     13.00    5000      4500      0         FULL
Z01  Municipal Hospital 4000     3000     11.80    4082      4000      0         FULL
Z02  Central Flood Shelter 3500     2500     11.60    100       96        3404     BELOW MIN
Z05  School Emergency Shelter 2600     1600     11.50    0          0         2600     UNSERVED
Z06  Fire and Emergency Service 2000     1500     10.90    0          0         2000     UNSERVED
=====

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.00)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water      : 20400.00 L
Total Allocated Water      : 13500.00 L
Total Effective Water Delivered : 12396.18 L
Total Unresolved Requirement : 8003.82 L
Remaining Water            : 0.00 L

Service Conditions
-----
Full Requirement Satisfied : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum              : 1 zone(s)
Completely Unserved        : 2 zone(s)
```

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Enter your choice: 6

===== CASE 6 =====
Previously underserved zones from Case 1:
Z02 - Central Flood Shelter
Z05 - School Emergency Shelter
Z06 - Fire and Emergency Service

Enter Zone ID from the underserved list: Z02
Current waiting cycles: 1
Enter new waiting cycles: 3

Waiting cycles of Z02 changed to 3.

=====
ALLOCATION RESULT
=====
ID   Zone           Request  Minimum  Priority  Allocated  Delivered  Shortage  Status
-----
Z02  Central Flood Shelter 3500     2500     15.60    3646      3500      0         FULL
Z04  Ward 5           3800     1800     13.80    4318      3800      0         MINIMUM
Z03  Ward 3           4500     2000     13.20    4891      4500      0         FULL
Z01  Municipal Hospital 4000     3000     11.80    645       632       3368     BELOW MIN
Z05  School Emergency Shelter 2600     1600     11.50    0          0         2600     UNSERVED
Z06  Fire and Emergency Service 2000     1500     10.90    0          0         2000     UNSERVED
=====

Priority Order:
1. Z02 - Central Flood Shelter (Priority 15.60)
2. Z04 - Ward 5 (Priority 13.80)
3. Z03 - Ward 3 (Priority 13.20)
4. Z01 - Municipal Hospital (Priority 11.80)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water      : 20400.00 L
Total Allocated Water      : 13500.00 L
Total Effective Water Delivered : 12431.79 L
Total Unresolved Requirement : 7968.21 L
Remaining Water            : 0.00 L

Service Conditions
-----
Full Requirement Satisfied : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
```

Enter your choice: 7

===== CASE 7 =====

Independent Test Case

Enter any available water quantity: 6000

ALLOCATION RESULT								
ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.20	1682	1547	2953	BELOW MIN
Z01	Municipal Hospital	4000	3000	11.80	0	0	4000	UNSERVED
Z02	Central Flood Shelter	3500	2500	11.60	0	0	3500	UNSERVED
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

Priority Order:

1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary

Total Requested Water : 20400.00 L
Total Allocated Water : 6000.00 L
Total Effective Water Delivered : 5347.27 L
Total Unresolved Requirement : 15052.73 L
Remaining Water : 0.00 L

Service Conditions

Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum : 1 zone(s)
Completely Unserved : 4 zone(s)

Enter your choice: 8

Enter Zone ID: Z01

Zone Found

Zone ID : Z01
Service Zone : Municipal Hospital
Type : Hospital
Requested : 4000.00 L
Minimum : 3000.00 L
Loss : 2.00 %
Waiting Cycles : 0
Priority : 11.80
Allocated : 4081.63 L
Loss Amount : 81.63 L
Effective Water : 4000.00 L
Shortage : 0.00 L
Condition : FULL

Enter your choice: 9

Highest Unresolved Requirement

Zone ID : Z02
Zone : Central Flood Shelter
Shortage : 3299.47 L

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EMERGENCY POTABLE WATER ALLOCATION

=====

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2. Case 2 - Enough Water
3. Case 3 - Below Minimum Requirement
4. Case 4 - Equal Priority
5. Case 5 - Increased Distribution Loss
6. Case 6 - Increased Waiting Period
7. Case 7 - Independent Test
8. Search a Zone
9. Find Highest Unresolved Requirement
10. Exit